

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
COMPUTER SCIENCE AND BUSINESS SYSTEMS

VII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
22PC1CB401	Usability Design of Software Applications	2	1	0	3	3
22HS1MG401	Financial Management	3	0	0	3	3
	PROFESSIONAL ELECTIVE – III					
22PE1CB401	Cognitive Science and Analytics	3	0	0	3	3
22PE1CB402	Generative AI					
22PE1CB403	Cryptology					
	PROFESSIONAL ELECTIVE – IV					
22PE1CB404	Quantum Computation and Quantum Information	3	0	0	3	3
22PE1CB405	Advanced Social, Text and Media Analytics					
22PE1CB406	Information Retrieval Systems					
	OPEN ELECTIVE – I					
22OE1CB401	Human Resource Management	3	0	0	3	3
22OE1CB402	Blockchain Technologies					
22OE1CB403	Information Security					
22PC2CB401	Usability Design of Software Applications Laboratory	0	0	2	2	1
22PC2AM402	Devops Laboratory	0	0	2	2	1
22PW4CB401	Major Project Phase – I	0	0	6	6	3
Total		14	1	10	25	20

L – Lecture T – Tutorial P – Practical D – Drawing CH – Contact Hours/Week
 C – Credits SE – Sessional Examination CA – Class Assessment ELA – Experiential Learning Assessment
 SEE – Semester End Examination D-D – Day to Day Evaluation LR – Lab Record
 CP – Course Project PE – Practical Examination

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
COMPUTER SCIENCE AND BUSINESS SYSTEMS

VIII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
	PROFESSIONAL ELECTIVE – V					
22PE1CB407	Behavioural Economics	3	0	0	3	3
22PE1CB408	Computational Finance and Modeling					
22PE1CB409	Psychology					
	PROFESSIONAL ELECTIVE – VI					
22PE1CB410	Enterprise Systems	3	0	0	3	3
22PE1CB411	Advance Finance					
22PE1CB412	Image Processing and Pattern Recognition					
	OPEN ELECTIVE – II					
22OE1CB404	IT Project Management	3	0	0	3	3
22OE1CB405	Services Science and Service Operational Management					
22OE1CB406	Marketing Research and Marketing Management					
22PW4CB402	Major Project Phase – II	0	0	22	22	11
Total		9	0	22	31	20

L – Lecture T – Tutorial P – Practical D – Drawing
C – Credits SE – Sessional Examination CA – Class Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation
CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
ELA – Experiential Learning Assessment
LR – Lab Record

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1CB401) USABILITY DESIGN OF SOFTWARE APPLICATIONS

TEACHING SCHEME		
L	T/P	C
2	1	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To create a learning system through which management students can enhance their innovation and creative thinking skills
- To acquaint themselves with the special challenges of starting new ventures
- To use IPR as an effective tool to protect their innovations and intangible assets from exploitation
- To gain expertise in redesigning an existing application or website for better user experience

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Demonstrate understanding of User-Centered Design principles and heuristic evaluation

CO-2: Apply design lifecycle phases and usability testing for user-centered development

CO-3: Utilize UX research tools and techniques to gather actionable insights

CO-4: Create personas, scenarios, and innovative designs using design thinking methods

CO-5: Develop interactive prototypes using concept development and prototyping techniques

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	1	-	-	-	1	2	3	1	2	2	2	-
CO-2	2	3	3	2	2	-	-	2	1	2	3	2	1	2	-
CO-3	1	3	2	3	2	-	-	1	2	2	1	3	2	3	-
CO-4	2	3	3	1	-	-	-	1	3	2	3	3	1	3	-
CO-5	2	2	2	2	3	-	-	1	2	1	3	3	2	3	-

UNIT-I:

Introduction to User Centered Design: Basics of User Centered Design. Cognitive Aspects of User Centered Design, Product Appreciation Assignment – Evaluating the product from user centered design aspects such as functionality, ease of use, ergonomics, aesthetics.

Heuristic Evaluation: 10 Heuristic Principles, Examples Heuristic Evaluation: Group Assignment initiation (Website and App) Evaluation for key tasks of the app or website for heuristic principles, severity, recommendations.

UNIT-II:

Project Design Lifecycle: Process of Interaction Design and phases.

Requirements: Types of requirements, Requirement discovery methods, Pilot studies.

Concrete Design: Visual appearance, Accessibility and national culture considerations.

Prototyping: Types and purposes, Story boarding from scenarios, Card-based prototypes from use cases, Steps for prototyping.

Usability Testing: Importance and goals, identifying problems, improving design, Learning user behaviour, Types of usability testing, Qualitative vs. Quantitative, Remote vs. In-person, Usability case study.

UNIT-III:

UX Research: Understanding users, their goals, context of use, and environment of use, Importance of gathering user attitudes, perceptions, and behaviors.

Research tools: Surveys, Interviews, Focus groups, Usability tests, Field studies, Research Techniques.

Contextual Inquiry: Semi-structured interviews combined with observation. Conducted in the user's environment for realistic insights.

Interviews: Structured, Semi-structured, Unstructured, Focus Groups, Questionnaires. Competitive Analysis, SWOT Analysis, Context of Use Analysis.

UNIT-IV:

Personas and Scenarios: Scenarios and Persona Technique.

Overview of Design Thinking Technique: Design thinking, Lean UX, Agile UX, Brainstorming for Innovation.

UNIT-V:

Development and Prototyping: Concept Development, Task Flow Detailing, Prototyping: Low-Fidelity Prototypes: Paper sketches, storyboarding, clickable wireframes, Wizard of Oz simulations.

High-Fidelity Prototypes, Prototyping Tools. Prototyping Techniques: Clickable Wireframes, Storyboarding, Sketching, Wizard of Oz. Construction: Physical Computing, SDKs: Software Development Kits

TEXT BOOKS:

1. Interaction Design: Beyond Human-Computer Interaction, Jenny Preece, Helen Sharp and Yvonne Rogers, 4th Edition, Wiley, 2015
2. Observing the User Experience: A Practitioner's Guide to User Research, Elizabeth Goodman, Mike Kuniavsky, Andrea Moed, 2nd Edition, Morgan Kaufmann, 2012

REFERENCES:

1. About Face, Alan Cooper and Robert Reimann, 4th Edition, Wiley, 2014
2. The Elements of User Experience: User-Centered Design for the Web and Beyond, Jesse James Garrett, 2nd Edition, New Riders, 2010
3. Understanding Design Thinking, Lean, and Agile, Jonny Schneider, O'Reilly, 2017
4. The Design of Everyday Things, Don Norman

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22HS1MG401) FINANCIAL MANAGEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the fundamentals of financial management and the time value of money
- To explain the valuation of securities and calculation of portfolio of risk and return
- To understand the fundamentals of leverage and how resources are used efficiently, effectively, and economically
- To understand the need for working capital management and its estimation
- To understand cash management and accounts receivables in business firms

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyze the concepts of financial management, the importance of the Time value of Money, and the valuation of Securities

CO-2: Apply concepts of risk and return such as cost of capital, risk, and return, and Financial Leverage

CO-3: Evaluate investment proposals through capital budgeting

CO-4: Estimation of working capital requirements and working capital management

CO-5: Apply the concepts of cash management of accounts receivables

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	-	-	-	-	-	-	-	1	2	3	1	-	-	-	-
CO-2	-	-	-	-	-	-	-	2	2	3	2	-	-	-	-
CO-3	-	-	-	-	-	-	-	1	3	2	1	-	-	-	-
CO-4	-	-	-	-	-	-	-	1	3	2	3	-	-	-	-
CO-5	-	-	-	-	-	-	-	1	2	1	3	-	-	-	-

UNIT-I:

Introduction: Introduction to Financial Management - Goals of the firm - Financial Environments. Time Value of Money: Simple and Compound Interest Rates, Amortization, Computing more than once a year, Annuity Factor.

UNIT-II:

Valuation of Securities: Bond Valuation Preferred Stock Valuation, Common Stock Valuation, Concept of Yield and YTM. Risk & Return: Defining Risk and Return, Using Probability Distributions to Measure Risk, Attitudes Toward Risk, Risk and Return in a Portfolio Context, Diversification, The Capital Asset Pricing Model (CAPM)

UNIT-III:

Operating & Financial Leverage: Operating Leverage, Financial Leverage, Total Leverage,

Cost of Capital: Concept, Computation of Specific Cost of Capital for Equity - Preference – Debt, Weighted Average Cost of Capital – Factors affecting Cost of Capital.

UNIT-IV:

Capital Budgeting: The Capital Budgeting Concept & Process - An Overview, Capital Budgeting Techniques, Project Evaluation and Selection – simplify problems

UNIT-V:

Working Capital Management & Cash Management: Overview, and types of working capital, determinants of working capital; estimation of working capital.

Motives for Holding cash, Speeding Up Cash Receipts, Slowing Down Cash Payouts, Electronic Commerce, Factoring;

Accounts Receivable Management: Credit & Collection Policies

TEXT BOOKS:

1. Financial Management, Prasanna Chandra, Theory & Practice, Tata McGraw-Hill
2. Financial Management, M. Y. Khan, P. K. Jain, 8th Edition, McGraw-Hill, 2018
3. Fundamentals of Financial Management, Van Horne and Wachowicz, Prentice Hall/Pearson Education

REFERENCES:

1. Financial Management, Srivastava, Misra, OUP
2. Investment Analysis and Management, Charles P. Jones, 9th Edition, 2004
3. Financial Management, M. Pandey, Vikas Publishing House, 10th Edition, 2012
4. Financial Management (Theory, Problems, Cases, Ravi M. Kishore, 8th Edition, Taxmann, 2020
5. Fundamentals of Financial Management, Eugene F. Brigham, Joel F. Houston, 14th Edition, Cengage, 2021

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22OE1CB401) HUMAN RESOURCE MANAGEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the functions, systems, policies, and applications of Human Resource Management in organizations
- To familiarize how HRM assesses the constraints and opportunities associated with managing employees in different socio-economic and political contexts
- To provide better inputs on HR planning.
- To explain the emerging horizons of HRM in Organizations
- To comprehend how HRM activities lead to the performance and sustainability of the organization

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Analyze the concept of human resource management and its relevance in organizations

CO-2: Identify the necessary skill set for the application of various HR functions

CO-3: Design and implement HR Planning in organisations

CO-4: Analyze the strategic HR activities in real time organizational environment

CO-5: Assess cross-cultural work dynamics and HR activities

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	-	-	-	-	-	-	-	2	3	2	3	-	-	-	-
CO-2	-	-	-	-	2	-	-	2	3	2	3	-	-	-	-
CO-3	-	-	-	-	2	-	-	2	3	2	3	-	-	-	-
CO-4	-	-	-	-	-	3	-	2	3	2	3	-	-	-	-
CO-5	-	-	-	-	-	2	3	2	3	2	3	-	-	-	-

UNIT-I:

Human Resource Management: Concept, objectives and Challenges, HR Policies, Practices, Hr department, Line Management Responsibility in HRM.

UNIT-II:

Functional Areas of HRM1:

Human Resource Planning: Demand Forecasting– Action Plans, Retention-reasons and benefits, Redeployment-process, benefits, Succession Planning-Process.

Recruitment-Process, sources, Selection-process, benefits, and compensation-Purpose and importance, Employee relations-four methods for managing employee relations, training and development-methods, Performance appraisal-objectives-methods

UNIT-III:

Functional Areas of HRM2: Human resources accounting-objectives, Significance, HR audit-needs, Types, benefits of HR audit, human resource information systems (H.R.I.S.)-functions, importance, Payroll-Importance of Payroll Management, Payroll Management Process

UNIT-IV:

Strategic Management of Human Resources: SHRM- nature and Importance, relationship between HR strategy and overall corporate strategy, Managing Diversity in the Workplace-objectives and benefits

UNIT-V:

Human Resource Management in Service Sector- Special considerations for Service Sector including

- Managing the Customer – Employee Interaction
- Employee Empowerment and Customer Satisfaction
- Service Failure and Customer Recovery – the Role of Communication and Training
- Nature of Work for the Frontline Workers and the Backend
- Flexible Working Practices – Implications for HR

TEXT BOOKS:

1. Human Resource Management, Dessler G., Varrkey B., 16th Edition, Pearson Education, 2020
2. International Human Resource Management, Peter J. Dowling, Marion Festing, Allen D. Engle, Cengage, 2017
3. Human Resource and Personal Management, K. Aswathappa, 8th Edition, Tata McGraw-Hill, 2017

REFERENCES:

1. Human Resource and Personal Management, L. M. Prasad, Sultan Chand & Sons, 2017
2. Human Resource Management, Joseph J. Martocchio, 15th Edition, Pearson Education Champaign, 2019
3. Human Resource Management, Mathis R. L., Jackson J. H., 15th Edition, Jakarta, Salemba Empat
4. Human Resource Management, Decenzo D. A., 11th Edition, John Wiley
5. International Human Resource Management, B. Sebastian Reiche, Helene Tenzer, Anne-Wil Harzing, 6th Edition

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22OE1CB402) BLOCKCHAIN TECHNOLOGIES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To provide a comprehensive understanding of blockchain technology, its history, and its various applications
- To explore the structure and components of blockchain, including blocks, transactions, nodes, and smart contracts
- To study various consensus mechanisms and cryptographic techniques used in blockchain
- To differentiate between types of blockchains and their application areas
- To analyze real-world blockchain use cases, such as cryptocurrency, supply chain management, and real estate

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Explain the fundamental concepts of blockchain, including its characteristics, structure, and applications

CO-2: Analyze the architecture and components of blockchain technology, including different tiers and smart contracts

CO-3: Compare various consensus algorithms and cryptographic techniques used in blockchain systems

CO-4: Differentiate between various types of blockchains and evaluate their implementation using platforms like Hyperledger Fabric

CO-5: Apply blockchain technology to real-world case studies, including cryptocurrency, supply chain management, and land registration

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	1	1	1	1	1	1	2	2	1	3	2	2
CO-2	3	3	3	2	2	1	1	1	1	2	2	2	3	3	2
CO-3	3	3	3	3	2	1	1	1	1	2	2	2	3	3	2
CO-4	2	3	3	3	2	2	2	2	1	2	2	2	2	3	2
CO-5	2	3	3	3	3	3	3	2	2	3	3	3	2	3	3

UNIT-I:

Introduction to Blockchain: Introduction to centralized decentralized and distributed system, Distributed database, History of Blockchain, Various technical definitions of Blockchain, Generic elements of a blockchain: Block, Transaction, Peer-to-peer network, Node, Why It's Called "Blockchain", characteristics of Blockchain. Applications of blockchain technology.

UNIT-II:

Concept of Blockchain Technology, Structure of Blockchain, Component of block, genesis block, Tiers of blockchain technology Blockchain 0, Blockchain 1, Blockchain 2, Blockchain 3, Blockchain application framework, Smart Contract, why smart contract?, implement and deployment of smart contract,

UNIT-III:

Consensus mechanism, various consensus algorithm: Byzantine fault tolerance, proof of work (PoW), proof of stake (PoS), proof of authority (PoA), Cryptography and Technical Foundations, Confidentiality, Integrity, Authentication, CAP theorem and blockchain, Public and private keys, RSA, Discrete logarithm problem, Message Digest (MD), Hashing, Hash Function, Secure Hash Algorithms (SHAs), Design of Secure Hash Algorithms (SHA), SHA-256, Design of SHA3, Elliptic Curve Digital signature algorithm.

UNIT-IV:

Types of Blockchain: Public blockchains, Private blockchains, Semi-private blockchains, Tokenization of assets, Tokenized blockchains, Tokenless blockchains, ledger, Permissioned ledger Distributed ledger Shared ledger application area of various types of blockchain.

Implementation Platforms: Hyperledger as a protocol, Reference architecture, Hyperledger Fabric, Transaction Flow, Hyperledger Fabric Details, Fabric Membership, Fabric Membership, comparison between various platform.

UNIT-V:

Blockchain Case Study: Crypto Currency: Bitcoin, Bitcoin definition, Keys and addresses, Public keys in Bitcoin, Private, keys in Bitcoin, Bitcoin currency units, Financial markets and trading, Supply chain Management, Real Estate: Land registration process.

TEXT BOOKS:

1. Mastering Blockchain, Imanan Bashir
2. Blockchain Technology, Chandramouli Subramanian, Universities Press

REFERENCES:

1. Blockchain For Dummies®, IBM Limited Edition, John Wiley & Sons
2. Blockchain Basics: A Non-Technical Introduction in 25 Steps, Daniel Drescher
3. Blockchain Revolution, Don Tapscott & Alex Tapscott
4. Blockchain for Dummies, Tiana Laurence

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22OE1CB403) INFORMATION SECURITY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce key security parameters such as confidentiality, integrity, and availability
- To understand various access control models and their applications
- To explore different security policies and system design principles for secure computing
- To analyze logic-based security threats, including malicious logic and intrusion detection
- To examine security architectures in operating systems and databases

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Explain fundamental security concepts, including security policies, threats, and the security life cycle

CO-2: Compare different access control models and their role in system security

CO-3: Analyze security policies and system design principles to ensure confidentiality, integrity, and access control

CO-4: Identify and evaluate security threats, including malicious logic, intrusion detection, and auditing techniques

CO-5: Assess security mechanisms in operating systems and databases for enterprise security

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	2	1	2	2	1	2	2	2	2	3	2	1
CO-2	3	3	3	2	3	2	1	1	2	2	2	2	3	2	2
CO-3	3	3	3	2	3	2	2	1	2	3	3	3	3	3	2
CO-4	2	3	3	3	3	3	1	2	3	2	2	3	3	3	2
CO-5	3	3	3	3	3	3	2	3	2	3	2	3	3	3	3

UNIT-I:

Overview of Security Parameters: Confidentiality, integrity and availability; Security violation and threats; Security policy and procedure; Assumptions and Trust; Security Assurance, Implementation and Operational Issues; Security Life Cycle.

UNIT-II:

Access Control Models: Discretionary, mandatory, roll-based and task-based models, unified models, access control algebra, temporal and spatio-temporal models.

UNIT-III:

Security Policies: Confidentiality policies, integrity policies, hybrid policies, non-interference and policy composition, international standards.

Systems Design: Design principles, representing identity, control of access and information flow, confinement problem.

Assurance: Building systems with assurance, formal methods, evaluating systems.

UNIT-IV:

Logic-based System: Malicious logic, vulnerability analysis, auditing, intrusion detection.

Applications: Network security, operating system security, user security, program security.

Special Topics: Data privacy, introduction to digital forensics, enterprise security specification.

UNIT-V:

Operating Systems Security: Security Architecture, Analysis of Security in Linux/Windows.

Database Security: Security Architecture, Enterprise security, Database auditing.

TEXT BOOKS:

1. Security Engineering, Ross Anderson
2. Computer Security: Art and Science, M. Bishop, Pearson Education
3. Information Security: Principles and Practice, M. Stamp

REFERENCES:

1. Security in Computing, C. P. Pfleeger, S. L. Pfleeger, J. Margulies
2. Secure Programming HOW TO, David Wheeler
3. Browser Security Handbook, Michael Zalewski
4. Handbook of Database Security, M. Gertz, S. Jajodia

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1CB401) COGNITIVE SCIENCE AND ANALYTICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To identify core concepts of cognitive analytics for intelligent applications
- To explore language semantics and information processing models
- To focus on data engineering principles of cognitive science
- To understand the need for Perception in Real-world scenarios
- To apply Deep Learning principles for large databases

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Apply basic concepts of data analytics in the field of AI and cognitive engineering

CO-2: Explore the fundamentals of the evolution of cognitive science for language semantics, data interpretation and memory processing

CO-3: Demonstrate the concepts of cognitive analytics to develop an information processing system for business specifications

CO-4: Analyse deep understanding of learning patterns and cognitive engagement of virtual devices connected in a digital environment

CO-5: Apply deep learning concepts in the world dealing with unstructured forms

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	3	3	3	1	1	1	2	2	1	2	3	2	2
CO-2	3	3	3	3	3	3	2	1	2	2	1	2	1	2	2
CO-3	3	3	3	3	3	1	2	2	2	2	2	2	3	3	3
CO-4	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
CO-5	2	2	3	3	1	1	1	1	2	1	2	3	2	3	3

UNIT-I:

Foundational Areas of Analytics Cognitive Science:

Introduction to Analytics: Definition, Description & Evolution of Analytics, History of Analytics, and Applicability of Analytics with development of Technology and Computer, How Analytics entered mainstream

Concepts of Analytics: Various overlapping concepts and fields of Analytics such as Data Mining, Machine Learning, Artificial Intelligence and Simulation

Emerging Areas in Analytics: Understanding of emerging research areas of Analytics: Evolutionary computation, Simulation, Machine learning/data mining

Value Chain of Analytics: Descriptive Analytics Covering Exploratory Data Analysis & Basic of Statistics, Diagnostics Analytics: BI/Analysis, Trend, Pattern, Simultaneous Relationship.

Predictive Analytics: Cause-effect relationship and Futuristic prediction in terms of probabilities, Continuous & Categorical Predictions.

Introduction to the study of cognitive sciences, Brief history of cognitive science development and Methodological concerns in philosophy. Foundational areas of **Cognitive Psychology:** Key Concepts - Nativism, Empiricism, Dualism, Structuralism, Functionalism.

UNIT-II:

Core Concepts of Cognitive Science: Human Brain-Memory Processing

Introduction to Central Nervous System-Neural Representations, The Architecture of the Mind, Neuro Psychology, Understand Brain and Sensory Motor Information: Fundamentals of Neuro Science, Processing of sensory information in the brain, and Brain Imaging Elements. The Organization of the Mind.

Language & Linguistic Knowledge: Background and details of Syntax & Semantics, Understanding of Generative Linguistics.

Memory & Processing: Theory of Information Processing, Fundamentals of Short-term Memory, Computational Neuroscience

Cognitive Analytics Concepts: Gestalt Principles, Text Analytics, Learning Analytics, Data Mining, Cognitive Systems, Cognitive Computing, Learning, Data Science, Machine Learning, Big Data Analytics and Business Analytics.

UNIT-III:

Data Theory, Taxonomy of Data and Cognitive Analytics:

Data as a Whole: Understanding of Data as a whole for distinguishing and relating Various types of data and Categorization of Data: Structured, Unstructured Data, Quantitative & Qualitative Data.

Views of Data: Understanding Data as an interdisciplinary framework for learning methodologies: covering statistics, neural networks, fuzzy logic, Association Rule Mining, Sequential Pattern Mining

Various Types of Scales: Nominal, Ordinal, Interval & Ratio Scales

Multivariate Data Analytics: Predictive analytics techniques such as multiple linear regression, logistic regression, decision tree learning, Clustering, and forecasting techniques, Stochastic models, Hypothesis testing, analysis of variance, and correlation.

UNIT-IV:

Artificial Intelligence & Machine Learning in Cognitive World:

Fundamentals of Artificial Intelligence: Various areas of AI:

- a) **Knowledge:** Text Analytics, Topic Modelling, Natural Language Processing (NLP), Natural Language Generation (NLG), Natural Language Understanding (NLU), Named-entity recognition (NER), Role of Grammar in Language Processing, Connectionist Models, Physical symbol systems and language of thought-Applying the Symbolic Paradigm
- b) **Perception:** Cognitive Bias Problem, Is AI Biased? Image Analytics, Video Analytics & Audio Analytics
- c) **Memory:** Cognitive Engagement: BOTs, Virtual & Digital Assistants, Augmented Reality, Virtual Reality, Mixed Reality
- d) **Learning:** Intelligent Automation, Reinforcement Learning, Intelligent Agents

Spectrum of AI

- a) **Reactive Machine:** Low memory, works on Known rules, such as Object Detection/Games/Recommendations specific to known Rules

- b) **Limited Memory:** Memory used to learn and improve continuously such as Most ML Models, Automated Vehicles
- c) **Theory of Mind:** Machine Understands and responds such as BoTs/Virtual/Digital Assistants
- d) **Self-Aware:** Human-like intelligence such as Super Robots in Space etc.

UNIT-V:

Cognitive Solutions, Approach & Methodology:

World Standard Methodology: Deep Learning Modules

Real-Life Workaround Artificial Intelligence, Machine Learning and Deep Learning: A few Selected Commonly Used Techniques & Algorithms: ANN (Artificial Neural Network), CNN (Convolutional Neural Network), RNN (Recurrent Neural Network); RN Architecture: LSTM, Bidirectional LSTM, Gated Recurrent Unit (GRU), CTRNN (Continuous Time RNN) CNN Architectures: VGG16, Alexnet, InceptionNet, ResNet, Googlenet

Autoencoders: Denoising Autoencoder, GAN

Transformers: Attention based Encoder and Decoder: Eg- BERT (Bidirectional Encoder Representations from Transformers), Generative Pretrained Transformers GPT-3, GPT-2, BERT, XLNet, and Roberta, Explainable AI

TEXT BOOKS:

1. Cognitive Science: An Introduction, Neil Stillings, Steven E. Weisler, Christopher H. Chase and Mark H. Feinstein, 2nd Edition, 1995
2. Cognitive Science: An Introduction to the Science of the Mind, José Luis Bermúdez, Cambridge University Press, 2010
3. Computational Vision: Information Processing in Perception and Visual Behavior Hanspeter A. Mallot, Translated John S. Allen

REFERENCES:

1. How the Mind Works, Steven Pinker, 2009
2. Cognitive Science: An Interdisciplinary Approach, Carolyn Panzer Sobel and Paul Li, 2013
3. Mind: Introduction to Cognitive Science, Paul Thagard, 2nd Edition, MIT Press, 2005
4. Cognitive Science: An Introduction to the Study of Mind, J. Friedenberg and G. Silverman, 2006
5. Cognitive Psychology, Sternberg Robert J. and Karin Sternberg, Thomson/Wadsworth, 2006

B.Tech. VII Semester

(22PE1CB402) GENERATIVE AI

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce the foundations of Artificial Intelligence (AI) and Neural Networks, including key concepts like deep learning and backpropagation
- To explore deep learning techniques such as LSTM, Transformers, and generative models (GANs, VAEs, and diffusion models)
- To develop an understanding of text and image generation using generative AI models
- To analyze large language models (LLMs) such as BERT and their applications in NLP tasks
- To implement generative AI applications using Python and TensorFlow 2, including ethical considerations and emerging trends

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Explain the history, evolution, and fundamentals of AI, machine learning, and neural networks, including backpropagation and training processes

CO-2: Analyze deep learning models such as LSTMs, Transformers, and generative models (GANs, VAEs) and implement them for real-world applications

CO-3: Implement and fine-tune generative AI models for text and image generation, including LSTM-based and Transformer-based approaches

CO-4: Understand and apply Large Language Models (LLMs) like BERT, pre-training techniques, and advanced transformer architectures for NLP tasks

CO-5: Utilize Python and TensorFlow 2 for implementing generative AI applications, including AI-generated music, and explore ethical considerations in AI

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	2	2	-	-	-	-	-	-	3	3	2	2
CO-2	3	3	3	2	3	-	-	-	-	-	-	2	3	3	2
CO-3	2	3	3	3	3	-	-	-	3	-	-	2	3	2	3
CO-4	2	2	3	2	3	2	-	-	2	2	-	3	3	3	3
CO-5	2	2	3	2	3	-	-	-	3	3	-	2	3	3	2

UNIT-I:

Foundations of AI and Neural Networks: History and evolution of AI/ML, Deep learning revolution, Transfer learning, History of Neural Natural Language Processing, Structure of Artificial Neural Networks, Steps in Training an Artificial Neural Network, Parameters and Hyperparameters, Backpropagation.

UNIT-II:

Deep Learning Essentials: Long Short-Term Memory (LSTM), Transformers and their role in sequence modelling.

Generative Models: Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Transformer models (e.g., BERT, GPT-3), Diffusion models

Case Study: Implementing Autoencoders for simple image datasets (e.g., MNIST).

UNIT-III:

Text Generation with Generative AI: Introduction to Text Generation, LSTM-based Text Generation, Transformer-based Text Generation, Fine-Tuning Language Models, and Text Generation Applications

Image Generation with Generative AI: Introduction to Image Generation, Implementing GANs for Image Generation Training and Fine-Tuning GANs, Generating Images with VAEs, Advanced Techniques in Image Generation, and Image and Video Generation Applications.

UNIT-IV:

Introduction to Large Language Models: Overview of Generative AI and Large Language Models. Basics of attention mechanisms and Transformer architecture. Pre-training techniques and transfer learning strategies.

BERT and Advanced Techniques: Understanding BERT architecture and pre-training objectives. Fine-tuning BERT for downstream NLP tasks. Exploration of advanced Transformer architectures and techniques

UNIT-V:

Python and TensorFlow 2 in Generative AI: Overview of Python and TensorFlow 2, Preprocessing and cleaning data for Generative AI applications. Visualizing data distributions and patterns in Generative AI datasets. Introduction to TensorFlow's computation graph and eager execution.

Advanced Techniques and Applications: Transfer learning in music generation, Fine-tuning generative models for specific music genres or styles, Ethical considerations in AI-generated music, Future directions and emerging trends in AI-driven music composition

TEXT BOOKS:

1. Generative AI with Python and TensorFlow 2: Create Images, Text, and Music with VAEs, GANs, LSTMs, Transformer Models, Joseph Babcock and Raghav Bali, 2024
2. Generative AI for everyone: Understanding the Essentials and Applications of this Breakthrough Technology, Altaf Rehmani

REFERENCE BOOKS:

1. Generative Adversarial Networks Cookbook: Over 100 Recipes to Build Generative Models Using Python, TensorFlows and Keras, Josh Kalin
2. Generative AI in Software Development: Beyond the Limitations of Traditional Coding, Jesse Sprinter, 2024
3. Deep Learning, Ian Goodfellow, Yoshua Bengio, and Aaron Courville
4. Neural Networks and Deep Learning: A Textbook, Charu C. Aggarwal

B.Tech. VII Semester

(22PE1CB403) CRYPTOLOGY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce the fundamental concepts of cryptography, including number theory and elementary cryptosystems
- To explore symmetric key cryptosystems, including stream ciphers and block ciphers
- To understand cryptographic hashing, authentication mechanisms, and public key cryptosystems
- To analyze cryptographic applications in secure electronic transactions, key management, and zero-knowledge protocols
- To introduce post-quantum cryptography and its potential impact on cryptographic security

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Explain the foundational principles of cryptography, including elementary number theory, pseudo-random bit generation, and basic security mechanisms

CO-2: Analyze symmetric key cryptosystems, including stream ciphers and block ciphers along with their operational modes

CO-3: Evaluate cryptographic hashing techniques, authentication mechanisms, and public key cryptosystems such as RSA and Elliptic Curve Cryptography (ECC), including digital signatures

CO-4: Apply cryptographic principles to real-world security applications, including electronic commerce, secure electronic transactions (SET), key management, zero-knowledge protocols, and cryptology in contact tracing applications

CO-5: Understand the fundamentals of post-quantum cryptography, including lattice-based encryption (NTRU), code-based cryptography (McEliece), and hash-based cryptographic techniques

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	1	1	-	-	2	-	-	1	2	3	2	1
CO-2	3	3	1	2	2	-	-	2	-	-	2	2	3	2	1
CO-3	3	3	3	3	3	-	-	-	1	-	2	3	3	3	2
CO-4	3	3	3	3	3	2	-	-	2	-	3	3	3	3	2
CO-5	3	3	3	3	3	2	3	1	1	-	2	3	3	3	3

UNIT-I:

Introduction to Cryptography: Elementary number theory, Pseudo-random bit generation, Elementary cryptosystems.

Basic Mechanisms: Security Attacks, Services and Mechanisms.

UNIT-II:

Basic Symmetric Key Cryptosystems:

Stream Cipher: A5/1, Grain family, RC4, Salsa and ChaCha, HC128, SNOW family, ZUC.

Block Ciphers: DES, Triple DES, AES, Cipher Block Modes of Operations

UNIT-III:

Hashing: Authentication Requirements, Hash Functions, Secure Hash Functions (SHA-256).

Public Key Cryptosystems: RSA, Elliptical Curve Cryptography; Digital signatures

UNIT-IV:

Security Applications: Electronic commerce (anonymous cash, micro-payments), Secure Electronic Transaction (SET), Key management, Zero-knowledge protocols, Cryptology in Contact Tracing Applications, Issues related to Quantum Cryptanalysis

UNIT-V:

Introductory Topics in Post-Quantum Cryptography: Post-Quantum Cryptography, lattice-based cryptography-NTRU Encryption, code-based cryptography-McEliece Cryptosystem, hash-based cryptography

TEXT BOOKS:

1. Cryptography, Theory and Practice, D. R. Stinson, CRC Press
2. Handbook of Applied Cryptography, A. J. Menezes, P. C. van Oorschot and S. A. Vanstone, CRC Press
3. A Course in Number Theory and Cryptography, N. Koblitz, GTM, Springer

REFERENCES:

1. Cryptography and Network Security. W. Stallings, Prentice Hall
2. Security Engineering, R. Anderson, Wiley
3. RC4 Stream Cipher and Its Variants, G. Paul and S. Maitra, CRC Press, Taylor & Francis Group, Chapman & Hall Book, 2012
4. Design & Cryptanalysis of ZUC - A Stream Cipher in Mobile Telephony, C. S. Mukherjee, D. Roy, S. Maitra, Springer, 2020
5. Contact Tracing in Post-Covid World - A Cryptologic Approach, P. Chakraborty, S. Maitra, M. Nandi, S. Talnikar, Springer, 2020

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1CB404) QUANTUM COMPUTATION AND QUANTUM INFORMATION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce the fundamental concepts of quantum information-including quantum states, operators, measurements, entanglement, and basic protocols along with quantum gates and circuits
- To present the foundational quantum algorithms and discuss their impact on classical symmetric key cryptosystems
- To explore advanced quantum algorithms with an emphasis on Shor's algorithm and its implications for factorization and discrete logarithm-based classical public key cryptosystems
- To examine the design principles, challenges, and applications of Quantum True Random Number Generators, including issues related to achieving true randomness and commercial product implementations
- To introduce and analyze various quantum key distribution protocols including their variations, challenges like device independence, and real-world applications

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Define key quantum concepts and simulate basic quantum communication protocols using simple quantum circuits.

CO-2: Analyze and compare quantum algorithms with classical approaches and evaluate their implications on symmetric cryptography.

CO-3: Explain the mathematical underpinnings of Shor's algorithm and assess its potential to compromise classical public key cryptographic systems.

CO-4: Evaluate QTRNG designs and understand their role in ensuring secure randomness in practical applications.

CO-5: Compare QKD protocols, assess their security features, and evaluate practical issues in deploying QKD systems

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	2	2	1	1	1	1	2	2	1	3	2	1
CO-2	3	3	3	3	2	2	1	1	1	2	2	2	3	3	1
CO-3	3	3	3	3	3	2	2	2	1	2	3	2	3	3	2
CO-4	3	3	3	3	3	3	2	2	1	3	3	2	3	3	2
CO-5	3	3	3	3	3	3	2	3	2	3	3	3	3	3	2

UNIT-I:

Introduction to Quantum Information: States, Operators, Measurements, Quantum Entanglement: Quantum Teleportation, Super-dense coding, CHSH Game, Quantum gates and circuits

UNIT-II:

Quantum Algorithms Basics: Deutsch-Jozsa, Simon, Grover, Shor, Implication of Grover's and Simon's algorithms towards classical symmetric key cryptosystems

UNIT-III:

Quantum Algorithms Advanced: Implication of Shor's algorithm towards factorization and Discrete Logarithm based classical public key cryptosystems

UNIT-IV:

Quantum True Random Number Generators (QTRNG): Detailed design and issues of quantumness, Commercial products and applications

UNIT-V:

Quantum Key Distribution (QKD): BB84, Ekert, Semi-Quantum QKD protocols and their variations, Issues of Device Independence, Commercial products

Introductory Topics in Post-Quantum Cryptography: Is cryptography dead? A taste of post-quantum cryptography. Challenges in post-quantum cryptography. Quantum-resistant public-key cryptographic algorithms (any two) Refer to <https://csrc.nist.gov/projects/post-quantum-cryptography>. May discuss any two ciphers from this list.

TEXT BOOKS:

1. Quantum Computation and Quantum Information, M. A. Nielsen and I. L. Chuang, Cambridge University Press
2. Preskill Lecture Notes: <http://www.theory.caltech.edu/~preskill/ph229/>

REFERENCES:

1. An Introduction to Quantum Computing, P. Kaye, R. Laflamme, and M. Mosca, Oxford University Press
2. Quantum Computer Science, N. David Mermin, Cambridge University Press
3. Quantum Algorithms for Cryptographically Significant Boolean Functions - An IBMQ Experience, SAPV Tharmashastha, D. Bera, A. Maitra and S. Maitra, Springer 2020
4. Handbook of Applied Cryptography, A. J. Menezes, P. C. van Oorschot, and S. A. Vanstone, CRC Press

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1CB405) ADVANCED SOCIAL, TEXT AND MEDIA ANALYTICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To provide an overview of common text mining and social media data analytic activities
- To understand the complexities of predictive tasks
- To understand and apply Web analytics
- To solve complex real-world problems for Classification analysis and recommendation systems
- To study the unstructured raw data in real-world connecting through social media

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand Text Mining and carry out Pattern Discovery

CO-2: Perform sentiment prediction and analysis on real-world datasets

CO-3: Find state-of-the-art web mining tools and libraries on realistic data sets as a basis for business decisions and applications and explore the use of social network analysis to understand the growing connectivity and complexity in the world around us on different scales – ranging from small groups to the World Wide Web

CO-4: Apply Classification models and provide solutions through recommendation systems

CO-5: Perform social network analysis to identify important social actors, subgroups (i.e., clusters), and network properties in social media sites such as Twitter, Facebook, and YouTube

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	3	3	3	1	2	1	2	1	2	3	3	3	3
CO-2	3	3	3	3	3	1	1	1	2	2	2	3	3	3	3
CO-3	3	3	3	3	3	1	1	1	2	2	2	3	3	3	3
CO-4	3	3	3	3	3	1	1	1	1	2	2	2	3	2	1
CO-5	3	3	3	3	3	1	1	1	2	2	2	3	3	3	3

UNIT-I:

Text Mining: Introduction, Core text mining operations, Preprocessing techniques, Categorization, Clustering, Information extraction, Content Analysis.

UNIT-II:

Methods & Approaches for Prediction Analysis: Opinion Mining, Natural Language Processing; Naïve Bayes approach, Topic Detection; Simple Predictive Modeling; Sentiment Analysis; Sentiment Prediction.

UNIT-III:

Web Analytics: Web Mining, Web analytics tools, Clickstream analysis, A/B testing, online surveys; Web search and retrieval, Search engine optimization, Web crawling and Indexing, Ranking algorithms, Web traffic models.

UNIT-IV:

Classification and Recommendations: RNN, LSTM, Bi-LSTM, attention Mechanisms, Transformers, Context-based Recommendation Systems, Explainable AI, and Autonomous agents.

UNIT-V:

Social Media Analytics and Making Connections: Graphs and Matrices, Graph Mining using Apriori, Basic measures for individuals and networks. Information visualization; Link analysis. Random graphs and network evolution. Social contexts: Affiliation and identity; Social network analysis.

Home Assignments:

1. **Language Analysis:** Students are expected to analyze the language of a category of text (e.g., literary, academic, social media) of their selection. Based on the analysis, students are expected to provide a critical description of the texts involved and possibly distinguish them from other texts and/or uncover relationships or concepts communicated by the text authors.
2. Students are required to Perform sentiment analysis using Twitter. Students will be required to use off-the-shelf software and/or code of their own to detect sentiment/emotion in the data and write a description of the methods they use and the results.

TEXT BOOKS:

1. The Text Mining Handbook: Advanced Approaches in Analyzing Unstructured Data, Ronen Feldman and James Sanger, Cambridge University Press, 2006
2. Analyzing Social Media Networks with NodeXL: Insights from a Connected World, Hansen, Derek, Ben Shneiderman, Marc Smith, Morgan Kaufmann, 2011
3. Web Analytics 2.0: The Art of Online Accountability, Avinash Kaushik, 2009

REFERENCES:

1. Introduction to Social Network Method Hanneman, Robert and Mark Riddle, 2005
2. Social Network Analysis: Methods and Applications, Wasserman S. & Faust K., Cambridge University Press, 1994
3. Theories of Communication Networks, Monge P. R. & Contractor N. S., 2003
4. <http://nosh.northwestern.edu/vita.html>, Oxford University Press
5. Web Data Mining Exploring Hyperlinks, Contents, and Usage Data, Bing Liu, 2nd Edition, Springer, 2011

B.Tech. VII Semester

(22PE1CB406) INFORMATION RETRIEVAL SYSTEMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce the fundamental concepts of information retrieval-including definitions, functional overview, and relationships with DBMS, digital libraries, and data warehouses
- To examine the processes involved in cataloguing and indexing, along with key data that support efficient information extraction
- To explore automatic indexing techniques-including statistical, natural language, and concept-based approaches-and introduce document clustering and thesaurus generation
- To investigate advanced search techniques such as search query formulation, similarity measures, ranking, relevance feedback, and selective dissemination in Boolean and hypertext systems
- To understand and analyze text search algorithms and evaluate information retrieval systems using established measurement criteria

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Explain the basic functionalities and interrelationships of information retrieval systems within the context of modern data management environments

CO-2: Design and implement cataloguing and indexing schemes using appropriate data structures to enhance retrieval performance

CO-3: Apply automatic indexing methods and clustering algorithms to organize and retrieve documents effectively from large datasets

CO-4: Formulate effective search queries and critically evaluate search results using various user search strategies and relevance measures

CO-5: Assess the performance of text search systems and conduct evaluations of information retrieval systems using standard metrics

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	1	2	1	1	1	1	2	1	2	3	2	2
CO-2	3	3	3	2	3	1	1	1	2	2	1	2	3	3	2
CO-3	3	3	2	3	3	1	1	1	2	2	1	2	3	3	2
CO-4	2	3	2	3	3	1	1	1	2	3	1	2	3	2	3
CO-5	3	3	3	3	3	1	1	1	2	3	2	3	3	2	2

UNIT-I:

Introduction: Definition, Objectives, Functional Overview, Relationship to DBMS, Digital libraries and Data Warehouses, Information Retrieval System Capabilities – Search, Browse, Miscellaneous.

UNIT-II:

Cataloging and Indexing: Objectives, Indexing Process, Automatic Indexing, Information Extraction

Data Structures: Introduction, Stemming Algorithms, Inverted file structures, N – gram data structure, PAT data structure, Signature file structure, Hypertext data structure.

UNIT-III:

Automatic Indexing: Classes of automatic indexing, Statistical Indexing, Natural language, Concept indexing, Hypertext linkages.

Document and Term Clustering: Introduction, Thesaurus generation, Item clustering, Hierarchy of clusters.

UNIT-IV:

User Search Techniques: Search statements and binding, Similarity measures and ranking, Relevance feedback, Selective dissemination of information search, weighted searches of Boolean Systems, Searching the Internet and hypertext.

UNIT-V:

Text Search Algorithms: Introduction, Software text search algorithms, Hardware text search systems.

Information System Evaluation: Introduction, Measures used in system evaluation, Measurement example – TREC results.

TEXT BOOKS:

1. Information Storage and Retrieval Systems: Theory and Implementation, Kowalski, Gerald, Mark T. Maybury, Springer
2. Modern Information Retrieval: Ricardo Baeza, Yates, Pearson Education, 2007

REFERENCES:

1. Information Retrieval: Algorithms and Heuristics, David A. Grossman and Ophir Frieder, 2nd Edition, Springer
2. Information Retrieval Data Structures and Algorithms, Frakes W. B., Ricardo Baeza-Yates, Prentice Hall, 1992

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2CB401) USABILITY DESIGN OF SOFTWARE APPLICATIONS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To create an application incorporating principles of user-centered design and usability
- To apply the design lifecycle, including personas, scenarios, prototyping, and usability testing, in application development
- To develop and validate interactive prototypes using low- and high-fidelity prototyping techniques
- To integrate modern tools and techniques, such as clickable wireframes and UX research, in iterative application development

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify and define a problem statement using user-centered design principles

CO-2: Apply the design lifecycle to analyze, design, and develop a user-centered application

CO-3: Develop low- and high-fidelity prototypes for identified scenarios using prototyping tools

CO-4: Conduct usability testing to evaluate the developed application

CO-5: Apply Testing to improve the application

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	2	1	1	-	-	2	2	1	3	3	2	1
CO-2	2	3	3	2	3	2	-	-	2	3	3	3	3	3	2
CO-3	3	2	2	3	3	-	-	1	2	2	3	3	3	3	-
CO-4	2	2	3	2	3	-	-	-	3	3	3	3	3	3	2
CO-5	2	2	3	2	3	-	-	-	3	3	3	3	3	3	2

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

WEEK 1, 2: Identify a website or app to redesign, focusing on heuristic evaluation principles and user-centered design justification.

WEEK 3, 4: Analyze the app or website using the design lifecycle: discovery, define, design, and usability testing phases.

WEEK 5, 6: Develop personas, scenarios and use-case for the chosen app or website.

WEEK 7, 8: Perform concept development and task flow detailing using techniques like 'Hero' Task Flows and user scenarios.

WEEK 9, 10: Create low and high-fidelity prototypes. Incorporate storyboarding.

WEEK 11, 12: Conduct usability testing with feedback collection, demonstrate the application, and provide improvements.

SOFTWARES: Rational Rose for use-case diagrams, Canva/Adobe for prototyping, persona, Scenario, story boarding and concept Development.

TEXT BOOKS:

1. Interaction Design: Beyond Human-Computer Interaction, Jenny Preece, Helen Sharp and Yvonne Rogers, 4th Edition
2. Observing the User Experience, A Practitioner's Guide to User Research, Elizabeth Goodman, Mike Kuniavsky, Andrea Moed, 2nd Edition

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2AM402) DEVOPS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Proficiency in at least one programming language (e.g., Python, Java, or Shell scripting), Basic understanding of REST APIs and software development concepts

COURSE OBJECTIVES:

- To provide hands-on experience in developing backend applications, implementing CRUD operations, and exposing REST APIs
- To introduce version control systems using Git and GitHub, enabling effective source code management and collaboration
- To familiarize with continuous integration and deployment (CI/CD) pipelines, using tools like CircleCI and GitHub Actions
- To equip with containerization skills, enabling them to package and manage applications using Docker and Kubernetes
- To develop proficiency in API testing and automation, utilizing tools like Postman and writing test cases for backend services

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Develop backend applications with CRUD operations and deploy them following DevOps best practices

CO-2: Manage and collaborate on source code using Git and GitHub, demonstrating version control strategies

CO-3: Set up and utilize CI/CD pipelines using CircleCI/GitHub Actions to automate builds and deployments

CO-4: Containerize applications using Docker, automate deployments, and orchestrate them using Kubernetes

CO-5: Perform API testing with Postman and write automated test cases to validate backend functionality

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	2	1	1	2	2	2	3	3	2	3
CO-2	3	2	2	2	3	1	1	1	3	2	2	2	3	2	2
CO-3	3	2	3	3	3	2	2	1	2	2	2	3	3	3	3
CO-4	3	3	3	3	3	2	2	1	2	2	2	3	3	3	3
CO-5	3	3	3	3	3	1	1	1	1	2	2	3	3	2	3

LIST OF PROGRAMS / EXPERIMENTS / EXERCISES:

1. Write code for a simple backend project. It should have basic CRUD operations and APIs (GET, POST, DELETE, PATCH/UPDATE).
2. Explore Git and GitHub commands.
3. Practice Source code management on GitHub. Experiment with the source code written in exercise 1.
4. Explore Docker commands for container management.
5. Develop a simple containerized application using Docker.
6. Integrate Kubernetes and Docker
7. Automate the process of running containerized application developed in exercise 7 using Kubernetes.
8. CircleCI/ Github Actions setup, explore the environment.
9. Demonstrate continuous integration and development using CircleCi/Github Actions
10. Install postman and explore manual API testing.
11. Develop test cases for simple APIs in the language of choice.
12. Integrate any of the suitable monitoring and logging dashboards with your application.
13. Hands-on parallel exploration of Terraform, and its configurations

TEXT BOOKS:

1. Practical Devops, Joakim Verona, 2nd Edition, 2018
2. DevOps Tools from Practitioner's Viewpoint, Deepak Gaikwad, Viral Thakkar, Wiley

REFERENCES:

1. DevOps: A Software Architect's Perspective, Len Bass, Ingo Weber, Liming Zhu, Addison Wesley
2. Edureka DevOps Full Course - https://youtu.be/S_0q75eD8Yc

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22OE1CB404) IT PROJECT MANAGEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand IT project management concepts, project overview and feasibility studies
- To apply project cost control and scheduling techniques like PERT and CPM
- To describe agile project management, principles and methodologies
- To be familiar with agile methodologies and techniques like scrum, devops, etc.

COURSE OUTCOMES: After completion of this course, the student should be able to

CO-1: Understand project management concepts, lifecycle phases, feasibility studies, and cost estimation

CO-2: Apply CPM and PERT techniques for project scheduling and cost optimization

CO-3: Manage project risks and apply control methods for successful execution

CO-4: Utilize agile methodologies like Scrum, Lean, and DevOps for adaptive project management

CO-5: Implement Scrum frameworks to improve team collaboration and productivity

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	1	-	-	-	1	2	3	1	2	2	2	-
CO-2	2	3	3	2	2	-	-	2	1	2	3	2	1	2	-
CO-3	1	3	2	3	2	-	-	1	2	2	1	3	2	3	-
CO-4	2	3	3	1	-	-	-	1	3	2	3	3	1	3	-
CO-5	2	2	2	2	3	-	-	1	2	1	3	3	2	3	-

UNIT-I:

Project Overview: Characteristics of a Project, Types of Projects- Development Projects, Research Projects, Scale of Projects, Project Management.

IT Project Management Framework: Components of the Framework- Managing the Project Lifecycle, Documenting the Project, Managing Project Documents and Artifacts, Project Management Lifecycle.

Identification Phase: Project Proposal-Cost Estimation, High-Level Time Frame, Risk Identification, Role of the Project Sponsor.

Initiation Phase: Assignment of a Project Manager, Development of the Project Charter, Authorization of Resources.

Planning Phase: Work Planning, Schedule Planning, Budget and Resource Planning, Execution and Control Phase: Development of Deliverables, User Acceptance and Approval.

Closeout Phase: Documentation of Lessons Learned, Development of a Project Closeout Report, Transfer of Deliverables.

Project Identification and Formulation: Opportunity Studies, Pre-feasibility and Feasibility Studies, Identification of Investment Opportunities.

Feasibility Study: Components of a Feasibility Study-Market and Demand Analysis, Technical Analysis, Financial Analysis, Economic and Environmental Benefits, Risk and Uncertainty.

Market and Demand Analysis: Steps in Market Analysis-Situational Analysis, Collection of Secondary Information, Market Surveys Demand Forecasting, Market Planning.

Cost Estimation: Importance of Cost Estimation, Types of Costs-Direct Costs, Indirect Costs.

Estimation Techniques: Analogous Estimation, Parametric Estimation, Bottom-Up Estimation, Three-Point Estimation, Vendor Analysis.

Project Appraisal: Market Appraisal, Technical Appraisal, Environmental Appraisal, Financial Appraisal, Economic Appraisal, Managerial Appraisal, Social Cost-Benefit Analysis

Detailed Project Report (DPR): Components of a DPR-Design Drawings, Cost Estimates, Specifications, Blueprint for Implementation.

UNIT-II:

Project Scheduling, Project Planning: Activities Identification, Sequential Ordering of Activities, Assigning Realistic Time.

Planning and Scheduling Techniques: Bar Chart, Network Analysis Techniques.

Network System: Components of a Network System- Activity, Event, Dummy Activity, Rules for Drawing Network Diagrams, Common Errors in Network Diagrams- Dangling, Looping or Cycling, Redundancy.

Critical Path Method (CPM): Characteristics of CPM, How to Draw a CPM Network, Critical Path Calculation, Forward Pass, Backward Pass, Float and Slack Time.

project Evaluation and Review Technique (PERT): Characteristics of PERT, Time Estimation (Optimistic, Pessimistic, Most Likely), Expected Time Calculation, Variance and Standard Deviation.

Time-Cost Trade-Off: Project Time-Cost Relationship, Crashing Activities, Steps to Shorten Project Duration.

Resource Allocation and Leveling: Resource Smoothing, Resource Leveling Techniques, Benefits of Resource Leveling.

UNIT-III:

Risk Analysis: Project Risks, Business Risks.

Risk Management: Identifying Risks, Assessing Risks, Prioritizing Risks, Quantitative Risk Assessment, Qualitative Risk Assessment, Risk Exposure and Risk Leverage.

Deciding Appropriate Actions: Accepting the Risk, Avoiding the Risk, Reducing the Risk, Transferring the Risk, Contingency Planning.

Risk Monitoring and Control: Risk Register and Risk Log, Planning, Monitoring, and Control Cycle.

Project Control: Monitoring Progress, Applying Corrective Actions, Types of Reporting Meetings: Team Meetings, Sponsor Meetings, Gantt Chart Representation of Progress.

Project Audit: General Audit, Detailed Audit, Technical Audit, Project Audit Recommendations, Ancillary Goals in Project Audit.

Project Termination: Termination by Extinction, Termination by Addition, Termination by Integration, Termination by Starvation, Termination Process and Implementation.

Final Project Report: Components of the Final Report- Project Performance, Administrative Performance, Organizational Structure, Project and Administrative Teams, Techniques of Project Management.

UNIT-IV:

Agile Project Management, Agile Principles, Agile Methodologies, Relationship between Agile, Scrum, Lean, DevOps, and IT, Agile vs. Waterfall Project Management, Agile Process, Agile Principles

Agile Methodologies: Extreme Programming (XP), Kanban, Lean Software Development, Scrum, Crystal, Dynamic Systems Development Method (DSDM), Feature-Driven Development (FDD).

SCRUM: Scrum Framework, Scrum Roles, Scrum Artifacts, Scrum Events.

Kanban, Lean Software Development, IT Service Management (ITSM), ITIL Framework, Adaptive Software Development (ASD).

UNIT-V:

SCRUM: Introduction to Scrum, Agile Overview, Agile Processes, Scrum Framework, Scrum Roles, Scrum Team-Scrum Master, Product Owner, Development Team. Development Activities in Scrum, Scrum Terminologies, Best Practices in Scrum, Scrum and Continuous Improvement, How Scrum Works.

TEXT BOOKS:

1. Succeeding with Agile: Software Development Using Scrum, Mike Cohn
2. Project Management for IT Related Projects. ISEB Foundation, BCS Publications
3. Project Planning and Management with CPM and PERT, Kundan Singh and Mitthan Lal Kansal

REFERENCES:

1. Agile Project Management: 2 Books in 1: Beginner's Guide & Methodology, The Definitive Guide to Master Scrum, Kanban, XP, Crystal, FDD, DSDM
2. Scrum Guide- Scrum Masters, Ken Schwaber and Jeff Sutherland
3. DevOps For Beginners, Joseph Joyner
4. Agile Product Management with Scrum, Roman Pichler
5. Agile Project Management with Scrum, Ken Schwaber, Microsoft Professional

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22OE1CB405) SERVICES SCIENCE AND SERVICE OPERATIONAL MANAGEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand concepts related to services, operations and goods
- To learn about different ways to design Services and assess them using Service qualities
- To understand various methods to operate and manage Service businesses
- To know how innovation can be approached from Services point of view

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Describe and distinguish concepts related to services, operations and goods

CO-2: Identify ways to design Services and evaluate them using Service qualities

CO-3: Relate how various methods can be used to operate and manage Service businesses

CO-4: Analyze about how innovation can be approached from Services point of view

CO-5: Apply in real time service-oriented approaches

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	1	1	1	1	1	1	2	2	1	2	2	3
CO-2	3	3	3	2	2	2	1	1	2	2	2	2	2	3	3
CO-3	3	3	3	3	2	2	2	2	2	2	2	2	2	3	3
CO-4	2	3	3	3	3	3	3	2	2	3	3	2	2	3	3
CO-5	2	3	3	3	3	3	3	3	3	3	3	3	2	3	3

UNIT-I:

Introduction: Introduction to the course, Introduction to service operations, Role of service in economy and society, Introduction to Indian service sector

Nature of Services and Service Encounters: Differences between services and operations, Service package, characteristics, various frameworks to design service operation system, Kind of service encounter, importance of encounters

Service-Dominant Logic: From Goods-Dominant logic to Service-Dominant logic, Value Co-creation

UNIT-II:

Service Strategy and Competitiveness: Development of Strategic Service Vision (SSV), Data Envelopment Analysis

New Service Development: NSD cycle, Service Blueprinting, Elements of service delivery system

UNIT-III:

Service Design: Customer Journey and Service Design, Design Thinking methods to aid Service Design

Locating Facilities and Designing Their Layout: models of facility locations (Huff's retail model), Role of service-scape in layout design

UNIT-IV:

Service Quality: SERVQUAL, Walk through Audit, Dimensions of Service quality & other quality tools

Service Guarantee & Service Recovery: How to provide Service guarantee? How to recover from Service failure?

Forecasting Demand for Services: A review of different types of forecasting methods for demand forecasting.

UNIT-V:

Managing Capacity and Demand: Strategies for matching capacity and demand, Psychology of waiting, Application of various tools used in managing waiting line in services.

Managing Facilitating Goods: Review of inventory models, Role of inventory in services
Managing Service Supply Relationship: Understanding the supply chain/hub of service, Strategies for managing suppliers of service

Vehicle Routing Problem: Managing after sales service, Understanding services that involve transportation of people and vehicle, Techniques for optimizing vehicle routes

TEXT BOOKS:

1. Fitzsimmons & Fitzsimmons, Service Management: Operations, Strategy, Information Technology, 7th Edition, McGraw-Hill
2. Services, marketing: Integrating Customer Focus Across the Firm, Wilson A., Zeithaml V. A., Bitner M. J., & Gremler D. D., McGraw-Hill, 2012

REFERENCES:

1. Services Marketing, Lovelock, C, 7th Edition, Pearson Education, 2011
2. Service Design for Business: A Practical Guide to Optimizing the Customer Experience, Reason Ben and Lovlie Lavrans, Pan Macmillan, 2016
3. Open Services Innovation: Rethinking Your Business to Grow and Compete in A New Era, Chesbrough, H. John Wiley, 2010

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22OE1CB406) MARKETING RESEARCH AND MARKETING MANAGEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the basic concepts of marketing and consumer behavior
- To explain the various concepts related to product management and to understand the importance of packing and branding
- To understand pricing methods, advertising, and distribution strategies
- To explain the various statistical tools and techniques for data analysis in Marketing Research
- To understand the concepts of Internet marketing and the fundamentals of business-to-business marketing strategy, CRM strategies

COURSE OUTCOMES: After completion of this course, the student should be able to

CO-1: Interpret the dynamics of marketing and analyze how its various components interact in the real world

CO-2: Apply the various product management strategies and design packing and branding for the products and services

CO-3: Comprehend the dynamics of marketing mix elements such as pricing, distribution, and promotion mix elements to leverage marketing concepts for effective decision-making

CO-4: Leverage marketing research concepts, and identify statistical tools and techniques in marketing research for effective decision-making

CO-5: Plan for internet marketing, and position and price B2B products and services

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	-	-	-	-	-	-	-	1	3	2	3	-	-	-	-
CO-2	-	-	-	-	-	-	-	1	3	2	3	-	-	-	-
CO-3	-	-	-	-	-	-	-	1	3	2	3	-	-	-	-
CO-4	-	-	-	-	-	-	-	1	3	2	3	-	-	-	-
CO-5	-	-	-	-	-	-	-	1	3	2	3	-	-	-	-

UNIT-I:

Marketing Concepts and Applications: Introduction to Marketing & Core Concepts, Marketing of Services, Importance of marketing in service sector.

Marketing Planning & Environment: Elements of Marketing Mix, Analyzing needs & trends in Environment - Macro, Economic, Political, Technical & Social

Understanding the Consumer: Determinants of consumer behavior, Factors influencing consumer behavior

Market Segmentation: Meaning & Concept, Basis of segmentation, selection of segments, Market Segmentation strategies, Target Marketing, Product Positioning

UNIT-II:

Product Management: Product Life cycle concept, New Product development & strategy, Stages in New Product development, Product decision and strategies, Branding & packaging

UNIT-III:

Pricing, Promotion and Distribution Strategy: Policies & Practices – Pricing Methods & Price Determination Policies. Marketing Communication – The promotion mix, Advertising & Publicity, 5 M's of Advertising Management. Marketing Channels, Retailing, Marketing Communication.

UNIT-IV:

Marketing Research: Introduction, Type of Market Research, Scope, Objectives & Limitations Marketing Research Techniques, Survey Questionnaire design & drafting, Pricing Research, Media Research, Qualitative Research

Data Analysis: Use of various statistical tools – Descriptive & Inference Statistics, Statistical Hypothesis Testing, Multivariate Analysis - Discriminant Analysis, Cluster Analysis, Segmenting and Positioning, Factor Analysis

UNIT-V:

Business to Business Marketing: Fundamental of business markets. Organizational buying process. Business buyer needs. Market and sales potential. Product in business markets. Price in business markets. Place in business markets. Promotion in business markets. Relationship, networks, and customer relationship management. Business to Business marketing strategy

Internet Marketing: Introduction to Internet Marketing. Mapping fundamental concepts of Marketing (7Ps, STP); Strategy and Planning for Internet Marketing

TEXT BOOKS:

1. Marketing Management, Philip Kotler & Keller Kevin, 4th Edition, Pearson Education, 2019
2. Marketing Research: An Applied Approach, Malhotra N. K., Nunan D. & Birks D. F., Pearson Education, 2019

REFERENCES:

1. Marketing Management: A Relationship Approach, Hollensen S., Pearson Education, 2019
2. Marketing Management, Deepak R., Kanthiah Alias, and S. Jeyakumar, Educreation Publishing, 2019
3. Marketing Research, David A. Aaker, V. Kumar, Robert P. Leone, George S. Day, 13th Edition, Wiley, 2018

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CB407) BEHAVIOURAL ECONOMICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the concepts of behavioural economics
- To impart knowledge on current ideas and concepts regarding decision-making in economics, particularly from a behavioural science perspective
- To explore key departures and the consequences of behaviour of firms, households, and other economic entities
- To provide an overview of how behavioural principles have been applied to economic problems
- To understand strategic choice with the help of game theory and Nash equilibrium

COURSE OUTCOMES: After completion of this course, the student should be able to

CO-1: Identify and evaluate evidence for systematic departures of economic behaviour from the Predictions of the neoclassical model, and psychological explanations for these anomalies

CO-2: Incorporate psychologically motivated assumptions into economic models, and interpret the implications of these assumptions

CO-3: Analyze the principles of game theory and the ability to interpret incentives of people in interpersonal interactions

CO-4: Apply behavioural principles to economic problems

CO-5: Comprehend strategic choice using game theory and Nash equilibrium

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	-	-	-	-	-	-	-	1	2	1	3	-	-	-	-
CO-2	-	-	-	-	-	-	-	1	2	1	3	-	-	-	-
CO-3	-	-	-	-	-	-	-	1	2	1	3	-	-	-	-
CO-4	-	-	-	-	-	-	-	1	2	1	3	-	-	-	-
CO-5	-	-	-	-	-	-	-	1	2	1	3	-	-	-	-

UNIT-I:

Introduction: The neoclassical/standard model and behavioral economics in contrast; historical background; behavioral economics and other social sciences; theory and evidence in the social sciences and in behavioral economics; applications – gains and losses, money illusion, charitable donation.

Basics of Choice Theory: Revisiting the neoclassical model; utility in economics and psychology; models of rationality; connections with evolutionary biology and cognitive neuroscience; policy analysis – consumption and addiction, environmental

protection, retail therapy; applications – pricing, valuation, public goods, choice anomalies

UNIT-II:

Beliefs, Heuristics and Biases: Revisiting rationality; causal aspects of irrationality; different kinds of biases and beliefs; self-evaluation and self-projection; inconsistent and biased beliefs; probability estimation; trading applications – trade in counterfeit goods, financial trading behavior, trade in memorabilia

UNIT-III:

Choice under Uncertainty: Background and expected utility theory; prospect theory and other theories; reference points; loss aversion; marginal utility; decision and probability weighting; applications – ownership and trade, income and consumption, performance in sports.

UNIT-IV:

Intertemporal Choice: Geometric discounting; preferences over time; anomalies of inter-temporal decisions; hyperbolic discounting; instantaneous utility; alternative concepts – future projection, mental accounts, heterogeneous selves, procedural choice; policy analysis – mobile calls, credit cards, organization of government; applications – consumption and savings, clubs and membership, consumption planning

UNIT-V:

Strategic Choice:

1. Review of game theory and Nash equilibrium – strategies, information, equilibrium in pure and mixed strategies, iterated games, bargaining, signaling, learning; applications – competitive sports, bargaining and negotiation, monopoly and market entry
2. Individual preferences; choice anomalies and inconsistencies; social preferences; altruism; fairness; reciprocity; trust; learning; communication; intention; demographic and cultural aspects; social norms; compliance and punishment; inequity aversion; policy analysis – norms and markets, labor markets, market clearing, public goods; applications – logic and knowledge, voluntary contribution, compensation design.

TEXT BOOKS:

1. An Introduction to Behavioral Economics, N. Wilkinson and M. Klaes, 3rd Edition, Palgrave MacMillan
2. Introduction to Behavioral Economics, David R. Just, Wiley
3. Behavioral Economics: The Basics, Anke Plagnol, Philip Corr, Taylor and Francis, 2023

REFERENCES:

1. Behavioural Economics: A Very Short Introduction (Very Short Introductions) Illustrated Edition, Oxford Press
2. Behavioural Economics, Edward Cartwright, 4th Edition, Taylor and Francis, 2022
3. Behavioural Economics, Mohamed Buheji, AuthorHouse, 2023
4. Behavioural Economics: Psychology, Neuroscience, and The Human Side of Economics, David Orrell (Hot Science), 2021
5. Behavioral Economic Theory, John Lok, 1st Edition, Notion Press, 2024

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CB408) COMPUTATIONAL FINANCE AND MODELING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand existing financial models in a quantitative and mathematical way
- To understand the quantitative tools to solve complex problems in the areas of portfolio management, risk management and financial engineering
- To explain the approaches required to calculate the price of options
- To identify the methods required to analyse information from financial data and trading systems
- To understand the Financial Returns, volatility estimation by using statistical tools

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyze and interpret financial data, mathematical foundations of finance and financial modeling

CO-2: Apply the skills and knowledge of financial markets and instruments, and application of tools and techniques of quantitative finance and option pricing models

CO-3: Evaluate and manage various types of financial risks, and design and test computational finance models

CO-4: Analyze the advance knowledge in designing, developing and testing of computational finance models

CO-5: Analyze the financial returns and hedging strategies

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	-	-	-	-	-	-	-	2	1	3	1	-	-	-	-
CO-2	-	-	-	-	-	-	-	3	2	1	2	-	-	-	-
CO-3	-	-	-	-	-	-	-	2	3	2	1	-	-	-	-
CO-4	-	-	-	-	-	-	-	1	2	3	2	-	-	-	-
CO-5	-	-	-	-	-	-	-	2	3	2	1	-	-	-	-

UNIT-I:

Numerical methods relevant to integration, differentiation and solving the partial differential equations of mathematical finance: examples of exact solutions including Black Scholes and its relatives, finite difference methods including algorithms and question of stability and convergence, treatment of near and far boundary conditions, the connection with binomial models, interest rate models, early exercise, and the corresponding free boundary problems, and a brief introduction to numerical methods for solving multi-factor models.

UNIT-II:

Black-Scholes framework: Black-Scholes PDE: simple European calls and puts; put-call parity. The PDE for pricing commodity and currency options. Discontinuous payoffs - Binary and Digital options. The Greeks: theta, delta, gamma, vega & rho and their role in hedging. The mathematics of early exercise - American options: perpetual calls and puts; optimal exercise strategy and the smooth pasting condition. Volatility considerations - actual, historical, and implied volatility; local vol and volatility surfaces.

Simulation including random variable generation, variance reduction methods and statistical analysis of simulation output. Pseudo random numbers, Linear congruential generator, Mersenne twister RNG. The use of Monte Carlo simulation in solving applied problems on derivative pricing discussed in the current finance literature. The technical topics addressed include importance sampling, Monte Carlo integration, Simulation of Random walk and approximations to diffusion processes, martingale control variables, stratification, and the estimation of the "Greeks."

UNIT-III:

Financial Products and Markets: Introduction to the financial markets and the products which are traded in them: Equities, indices, foreign exchange, and commodities. Options contracts and strategies for speculation and hedging.

UNIT-IV:

Application areas include the pricing of American options, pricing interest rate dependent claims, and credit risk. The use of importance sampling for Monte Carlo simulation of VaR for portfolios of options.

UNIT –V:

Statistical Analysis of Financial Returns: Fat-tailed and skewed distributions, outliers, stylized facts of volatility, implied volatility surface, and volatility estimation using high frequency data.

Copulas, Hedging in incomplete markets, American Options, Exotic options, Electronic trading, Jump Diffusion Processes, High-dimensional covariance matrices, Extreme value theory, Statistical Arbitrage.

TEXT BOOKS:

1. Tools for Computational Finance, R. Seydel, 2nd Edition, Springer-Verlag, New York, 2004
2. Monte Carlo Methods in Financial Engineering, P. Glasserman, Springer-Verlag, 2004
3. Option Valuation under Stochastic Volatility, A. Lewis, Finance Press, Newport Beach, California, 2000

REFERENCES:

1. Numerical Recipes in C: The Art of Scientific Computing, W. Press, S. Teukolsky, W. Vetterling and B. Flannery, Cambridge University Press, 1997
2. Efficient Methods for Valuing Interest Rate Derivatives, A. Pelsser, Springer-Verlag, 2000
3. Statistics and Data Analysis for Financial Engineering, D. Ruppert
4. Statistical Analysis of Financial Data in S-Plus, R. Carmona
5. Time Series: Applications to Finance, N. H. Chan

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CB409) PSYCHOLOGY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the content areas of industrial psychology and the application of psychological theory to organizational issues
- To describe employment law, job analysis, recruitment and selection, training, performance appraisal and discipline, employee motivation, and workplace safety
- To explain how the science of human behaviour is used to select, develop, and manage employees
- To understand and learn how organizations can create a supportive work environment
- To familiarize the role of leadership and organizational culture in organizational development

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyse the major content areas of Industrial Psychology applications (i.e., job analysis, recruitment, selection, employment law, training, performance management, and health/well-being issues in the workplace)

CO-2: Apply the statistical concepts in the context of making sound data-based personal decisions

CO-3: Assess the behaviour of individuals and groups in organizations in terms of the key factors that influence organizational behavior.

CO-4: Examine the Importance of IO Psychology in employee motivation, diversity, and Fairness in the organization.

CO-5: Analyze the impact of leadership practices and Organizational culture and development

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	-	-	-	-	-	3	-	2	1	-	3	2	-	-	-
CO-2	-	-	-	3	-	2	-	-	3	-	3	2	-	-	-
CO-3	-	-	-	-	-	3	2	-	3	-	3	1	-	-	-
CO-4	-	-	-	-	-	3	2	2	1	-	3	1	-	-	-
CO-5	-	-	-	-	-	3	3	2	3	-	3	3	-	-	-

UNIT-I:

Introduction: What is I/O Psychology? Research Methods, Statistics and Evidence-based Practice, Introduction & Legal Context of Industrial Psychology, Job Analysis & Competency Modeling, Job Evaluation, Job Design & Employee Well-Being.

UNIT-II:

Evaluating the Quality of Performance Measures: Identifying Criteria & Validating Tests and Measures, Screening Methods, Intensive Methods

UNIT-III:

Organizational Fairness and Diversity Management: Employee Motivation-theories, Satisfaction and Commitment-significance, Fairness and Diversity-objectives, importance.

UNIT-IV:

Organizational Behaviour: The Organization of Work Behaviour, Teams in Organizations -types and formation, Stress Management: Demands of Life and Work

UNIT-V:

Leadership and Organizational Development: Leadership in Organizational Climate, Culture, and Development

TEXT BOOKS:

1. Work in the 21st Century: An Introduction to Industrial and Organizational Psychology, 6th Edition, Jeffrey M. Conte, Frank J. Landy
2. Industrial and Organizational Psychology: Research and Practice, Paul E. Spector, 2016
3. Industrial/Organizational Psychology: An Applied Approach, Aamodt M., 8th Edition, Wadsworth Publishing Co., 2015

REFERENCES:

1. Industrial / Organizational Psychology: Understanding the Workplace, Paul E. Levy, 2019
2. Industrial-Organizational Psychology, Miner B. J., McGraw-Hill, 1992
3. Organisational Behaviour, Stephen P. Robbins
4. Psychology Applied to Work, Paul M. Muchinsky, 13th Edition
5. Work and Organizational Psychology: Topics In Applied Psychology, Cary L. Cooper, Sebastiaan Rothmann, 3rd Edition, Taylor & Francis

B.Tech. VIII Semester

(22PE1CB410) ENTERPRISE SYSTEMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To introduce the essential concepts of ERP involved in business processes
- To impart skills in the design and implementation of ERP architecture
- To impart skills in the implementation of ERP architecture
- To familiarize with various tools and technologies for developing ERP for large project

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify the design involved in simple web applications using MVC architecture, related tools and techniques

CO-2: Implementing Microsoft .NET framework

CO-3: Evaluate SOA, ERP models, design and implement CRM, SRM models

CO-4: Analyze interactive network and applications

CO-5: Develop model for ERP for large projects understanding both hardware and software requirements

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	2	2	-	-	2	2	2	2	2	2	1	2
CO-2	2	3	2	-	3	-	-	1	3	2	3	2	3	2	1
CO-3	2	1	1	1	1	-	-	-	2	2	2	2	2	2	1
CO-4	2	2	2	1	2	-	-	-	2	2	2	2	3	1	1
CO-5	2	2	2	-	2	-	-	1	3	2	3	2	2	1	2

UNIT-I:

Model - View - Control (MVC) Architecture: Overview of MVC -MVC method of software development in a 3-tier environment -Control (MVC) development in a 3-tier environment.

UNIT-II:

Tools and Technologies: Microsoft .NET framework, PHP, Ruby on Rails, JavaScript, Ajax and Overview of SAP and Oracle Applications

UNIT-III:

ERP Architecture and Generic Modules: Service Oriented Architecture (SOA) - Principles of loose coupling – encapsulation - Inter-operability - Enterprise Resource Planning (ERP) systems and their architecture - Generic ERP Modules: Finance, HR, Materials Management, Investment - Examples of Domain Specific Modules

UNIT-IV:

ERP Technologies: Business Process Reengineering - Decision Support System - On-Line Analytical Processing -Electronic Data Exchange - Customer Relationship Management (CRM) - Supplier Relationship Management (SRM)

ERP Networking & Security: Overview of MPLS - Virtual Private Networks (VPN) – Firewalls - Network monitoring and enforcement of policies - ERP Security Issues – Authentication – Authorisation - Access control – Roles - single-sign-on -Directory servers - Audit trails - Digital signatures – Encryption - review of IPSec - SSL

UNIT-V:**Software Architectures for Enterprise Systems:**

Software: Acquisition Process – Tendering - conditions of contract - Commercial off the shelf software (COTS) Implementations - Bespoke Implementations - Total cost of ownership - Issues on using Open source software or free software and Licensed software

Hardware Architectures for Enterprise Systems: Hardware: Servers –Storage area networks - Storage units - Back-up strategies - Local Area Network (LAN) technologies and products - Data Centres - Hardware Acquisition - Disaster Recovery

TEXT BOOKS:

1. Enterprise Resource Planning, Alexis Leon, 4th Edition, Tata McGraw-Hill, 2020
2. Enterprise Resource Planning and Supply Chain Management, Kurbel K. E., Springer, 2016

REFERENCES:

1. Enterprise Resource Planning - Fundamentals of Design and Implementation, Ganesh K., Sanjay M., Anbuudayasankar S. P., Sivakumar P., Springer, 2014
2. Enterprise Systems for Management, Luvai F. Motiwalla and Jeff Thompson, 2nd Edition, Pearson, 2011
3. Enterprise Systems for Management, Motiwalla L. F., & Thompson J., 2nd Edition, Pearson, 2012

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CB411) ADVANCE FINANCE

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the various financial sources and instruments, and the corporate dividend decisions
- To explain leasing and decisions involving leasing, organizational goals with optimum investment, and corporate restructuring
- To understand financial restructuring and working capital management in business firms
- To understand risk and return in the derivatives
- To understand designing a swap agreements

COURSE OUTCOMES: After completion of the course, the student should be able to

CO -1: Analyze the decisions involved in sources of finance and dividend decisions

CO-2: Interpretation of business information and application of financial theory in corporate investment decisions, and assess working capital management and leasing contracts

CO-3: Evaluate the corporate and financial restructuring

CO-4: Analyze the derivative markets

CO-5: Designing and financial swaps

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	-	-	-	-	-	-	-	1	2	3	1	-	-	-	-
CO-2	-	-	-	-	-	-	-	2	3	2	2	-	-	-	-
CO-3	-	-	-	-	-	-	-	2	1	3	1	-	-	-	-
CO-4	-	-	-	-	-	-	-	1	2	2	3	-	-	-	-
CO-5	-	-	-	-	-	-	-	2	2	3	2	-	-	-	-

UNIT-I:

Sources of Funds (including regulatory framework): Types of securities; Issuing the capital in market; Pricing of issue; Valuation of Stocks and bonds

UNIT-II:

Dividend Decisions: Traditional Approach, Dividend Relevance Model, Miller and Modigliani Model, Stability of Dividends, Forms of Dividends, Issue of bonus shares, Stock Split.

UNIT-III:

Evaluation of Lease Contracts: Lease Definition, types agreements

UNIT-IV:

Corporate Restructuring: Mergers and Acquisitions- Types of Mergers, Evaluation of Merger Proposal; Take-over; Amalgamation; Leverage buy-out; Management buy-out; Corporate Failure and Liquidation

UNIT-V:

Financial Restructuring: Working Capital Management and Introduction to derivatives:

Share Split; Consolidation; Cancellation of Paid-up Capital; Other Mechanisms Working Capital Planning; Monitoring and Control of Working Capital; Working Capital Financing; Managing the Components of Working Capital-Cash Management- Receivable Management- Inventory Management-Introduction to derivatives; Basics of Futures, Forwards, Options, Swaps; Interest rate Payoff Diagrams, Pricing of Futures, Put Call Parity, Option Pricing using Binomial Model and Black Scholes Model Use of Derivatives for Risk-Return Management- Credit Default Swaps

Home Assignment:

A case study materials book will be given to students. Students are required to meet in groups before coming to class and prepare the case for the day. Instructors may ask the student groups to present their analysis and findings to the class.

Further, the topic for class discussion will be mentioned beforehand and students should be prepared to discuss these topics in class. A few topics are mentioned below as examples. Instructor can add or change any topic as per requirement.

1. **Topic:** Historical perspectives of markets like major boom and busts, bull and bear cycles, major market crashes, bubbles
2. **Topic:** Major scams in the market, e.g. Satyam case

TEXT BOOKS:

1. Principles of Corporate Finance, Brealey, Myers and Allen, 11th Edition, McGraw-Hill, 2017
2. Investment Analysis And Portfolio Management, Prasanna Chandra, 4th Edition, Tata McGraw-Hill, 2013
3. Accounting for Management, S. N. Maheswari, S. K. Maheswari, 4th Edition, Vikas Publishing House, 2018

REFERENCES:

1. Elements of Corporate Finance, S. M. Maheswari, Sultan Chand, 2016
2. Financial Management Text and Cases, IM Pandey, Vikas, 2012
3. Prasanna Chandra, Financial Management - Theory & Practice, 2018
4. Brigham & Houston- Fundamentals of Financial Management, 2018
5. Advanced Corporate Finance, Chandrasekhar Krishnamurti, S. R. Vishwanath, PHI, 2020

Case Study Materials: To be distributed for class discussion

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1CB412) IMAGE PROCESSING AND PATTERN RECOGNITION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To impart fundamental knowledge in the area image processing
- To understand various filters and techniques available to process an image
- To provide knowledge of the concepts related to image analysis
- To learn the fundamentals of pattern recognition

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the basics of image formation, digitization and representation

CO-2: Learn various image transformations and filtering techniques

CO-3: Learn various feature extraction techniques

CO-4: Comprehend the techniques of Image segmentation for better image analysis

CO-5: Understand colour image processing and morphological image processing techniques

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	2	1	1	1	1	1	2	1	1	3	2	1
CO-2	3	3	3	2	2	1	1	1	1	2	1	1	3	2	1
CO-3	3	3	3	3	2	1	1	1	1	2	1	1	3	3	1
CO-4	3	3	3	3	2	2	1	1	1	2	1	1	3	3	1
CO-5	3	3	3	3	3	2	1	1	1	2	1	1	3	2	1

UNIT-I:

Introduction: Image processing systems and its applications. Basic image file formats
Image formation: Geometric and photometric models; Digitization - sampling, quantization; Image definition and its representation, neighborhood metrics.

UNIT-II:

Intensity Transformations and Spatial Filtering: Enhancement, contrast stretching, histogram specification, local contrast enhancement; Smoothing, linear and order statistic filtering, sharpening, spatial convolution, Gaussian smoothing, DoG, LoG.

UNIT-III:

Image/Object Features Extraction: Textural features - gray level co-occurrence matrix; Moments; Connected component analysis; Convex hull; Distance transform, medial axis transform, skeletonization/thinning, shape properties.

UNIT-IV:

Segmentation: Pixel classification; Grey level thresholding, global/local thresholding; Optimum thresholding - Bayes analysis, Otsu method; Derivative based edge detection operators, edge detection/linking, Canny edge detector; Region growing, split/merge techniques, line detection, Hough transform.

UNIT-V:

Colour Image Processing: Fundamentals of different colour models - RGB, CMY, HSI, YCbCr, Lab; False colour; Pseudo colour; Enhancement; Segmentation.
Morphological Filtering Basics: Dilation and Erosion Operators, Top Hat Filters

TEXT BOOKS:

1. Digital Image Processing, R. C. Gonzalez and R. E. Woods, Prentice Hall
2. Image Processing, Analysis, and Machine Vision, Milan Sonka, Vaclav Hlavac, Roger Boyle, 3rd Edition, Brooks/Cole
3. Image Processing and Pattern Recognition: Fundamentals and Techniques, Frank Y. Shih, Wiley

REFERENCES:

1. Image Processing: The Fundamentals. Maria Petrou and Panagiota Bosdogianni, John Wiley
2. Digital Image Processing, K. R. Castleman, Prentice Hall, Englewood Cliffs
3. Digital Image Processing, William K. Pratt, Wiley
4. Fundamentals of Digital Image Processing, Anil K. Jain, Pearson
5. Pattern Recognition and Machine Learning, Christopher M. Bishop, Springer